

Jason Hernandez Lopez

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SUMMARY

Hands-on with autonomous systems, focused on mechanical design and electromechanical integration. I've managed and maintained L4 robotic fleets, contributed to system-level design and repairs, and supported R&D through validation, testing, and documentation.

EXPERIENCE

Automation Design Intern - (Winter 2026)

NewHaptics

January 2026 - Present, Ann Arbor, MI

- Redesigning and automating a sample imaging system under the Head Mechanical Engineer; developing CAD models with focus on mechanism design, kinematics, and structural integrity.
- Prototyping and integrating electromechanical subsystems, selecting motors and sensors, and implementing embedded control to enable automated task execution.
- Conducting system-level testing on iterative models; analyzing performance to refine CAD designs, improve reliability, and resolve control-level issues.

Robotics Hardware Intern

Serve Robotics

May 2025 - August 2025, Miami, FL

- Scaled operational fleet from 15 → 30 in 4 weeks and to 75 by week 10 (~17% weekly growth) by leading root-cause analysis across motors, encoders, sensors, and PCBs using power tools, oscilloscopes, and CAN tools.
- Sustained ≥65% fleet availability in first 4 weeks and ~80% over next 8 weeks by fabricating, installing, and validating drivetrain reinforcements and mechanical retrofits in CAD (Miami site ranked #1 nationwide in deliveries/robot).
- Averaged 10–12 repairs/week, including wire- and PCB-level soldering, electronics fixes, and sensor replacements; documented procedures with logs, videos, and diagnostics in JIRA.

Robot Technician - (Site Lead)

Starship Technologies

January 2025 - April 2025, Detroit, MI

- Managed and operated a fleet of 5 L4 autonomous robots; performed hands-on electromechanical maintenance, diagnosed and resolved drivetrain, perception, power, and sensor issues; documented procedures in JIRA and Confluence.
- Validated chassis, radar, and thermal upgrades through repeated field trials; identified failure modes and logged defects with proposed solutions.
- Managed parts, tooling, and maintenance workflows specific to my fleet: organized 100+ spare parts and placed hardware orders to keep test schedules on track.

PROJECTS

- **LEONARDO** - Automated Drawing Robot · Path Planning · Image Processing · Embedded Control
- **RANGER** - Autonomous Mobile Robot · Encoder Odometry · Closed-Loop Control · Computer Vision
- **FORKRUNNER** - Remote-Controlled Robot · High-Torque Drivetrain · Lead-Screw Forklift · Machined Components
- **STRYKER** - 6-DOF Robotic Arm · Mechanism Design · Actuator Selection · Electronics Integration

EDUCATION

Bachelor of Science in Robotics

University of Michigan College of Engineering · Ann Arbor, MI · Expected May 2027 · GPA: 3.5

SKILLS

Technical: Root-Cause Analysis, Hardware Validation & Testing, PCB-Level Soldering, Electromechanical Repair, Wiring & Circuitry.

Design: Fusion 360, SolidWorks, Onshape, LTspice, DFM/DFA, GD&T, Assembly Modeling, Tolerance Stack-ups & Fits, Mechanical Drawing.

Mechanical: Rapid Prototyping (3D Printing & Machining), Mechanical Maintenance, Assembly & Integration, Power Tools.

Programming: MATLAB, C++, SQL, Python, Data Structures & Algorithms, Debugging, AI-Assisted Development & Prompting.
